Visual System

Images used: 15.8, 15.12, 15.13, 15.15, 15.14, 15.6, 15.9

I. Overview of retinogeniculocortical pathway

A. Optical system

B. Retina

C. Optic nerve

D. Chiasm

E. Optic tract

F. Lateral geniculate nucleus

G. Optic radiations

H. Visual cortex

II. Optic nerve (cranial nerve II)

A. Four sections

1. Intraocular

a. Ganglion cells travel from all over the retina to converge on the optic disc.

b. Physiologic cup

c. Blind spot

2. Intraorbital

    a. Once the optic nerve leaves the globe, it is surrounded by pia, arachnoid, and dura

    b. Clinical correlate: Papilledema

    c. Posterior to the globe, the fibers become myelinated by oligodendrocytes.

3. Intracanalicular

4. Intracranial

III. Optic chiasm

A. Neighboring structures

B. Arrangement of fibers

IV. Optic tract

A. Most ganglion cell fibers travel through the optic tract to the LGN

B. Some ganglion cell fibers leave the optic tract prior to reaching the LGN

1. Superior colliculi

2. Suprachiasmatic nuclei

3. Pretectum

V. Lateral geniculate nuclei

A. Two main types of ganglion cells

1. P cells (parvocellular pathway)

2. M cells (magnocellular pathway)

VI. Optic radiations

A. Parietal lobe

B. Temporal lobe

VII. Visual cortex

A. aka primary visual cortex, V1, or striate cortex

B. Calcarine fissure

C. Cuneus gyrus

D. Lingual gyrus

E. Macular fibers terminate in the most posterior part of the occipital lobe.

VIII. Clinical implications

A. Information from the right visual field will project to the left visual cortex, and information from the left visual field will project to the right visual cortex.

B. General visual field concepts

1. On the retina, the optic nerve is always more nasal to the macula

2. Aligning visual fields

3. Position of the blind spot

C. Optic disc

1. Papillomacular bundle

2. Horizontal raphe

3. Visual field characteristics

D. Optic nerve (intraorbital, intracanalicular, intracranial)

1. Visual field characteristics

E. Chiasm

1. Visual field characteristics

F. Optic tract

1. Visual field characteristics

G. Radiations

1. Visual field characteristics

H. Visual cortex

1. Visual field characteristics